UNIVERSITY OF WOLLONGONG AUSTRALIA

Tutorial 02

Laplace transforms, initial conditions and inverse methods

Preparation: Lecture 02 covers unilateral transforms and initial-value terms. Lecture 02A Partial Fractions is required self-study: work through its coefficient methods before attempting inverse-transform questions. It is not scheduled for detailed classroom teaching.

Submission credit: 1% of the course grade for submitted work; correctness is not marked.

Show essential steps, substitutions and units.

Question 1 Basic Laplace checks

Use unilateral transforms and show each essential step. Signals in (a), (c) and (d) are zero for t < 0. In (b), keep the given initial state and assume no jump at t = 0.

a) Standard pairs and linearity. Find the transforms of the two functions below.

$$f(t) = 2 + 3t, \qquad g(t) = 4e^{-2t}$$

b) Initial-value terms. For the smooth displacement below, write the transforms of its first and second derivatives in terms of X(s). Do not solve an ODE.

$$x(0^-) = 0.10\ m, \qquad x'(0^-) = 0.20\ m/s$$

c) Simple inversion. Find y(t) by separating the transform into two simple fractions. Then state the output at t = 0^+ and as t approaches infinity from your time response.

$$Y(s) = \frac{6}{s(s+3)}$$

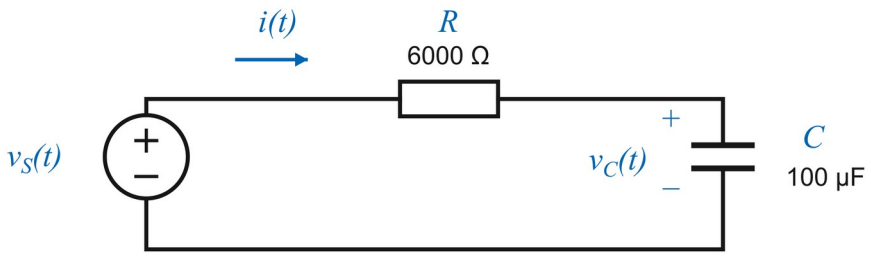
d) Check part (c) independently with the initial-value and final-value theorems. Check the final-value theorem condition before using it, and compare both results with y(t).

Question 2 RC charging from two initial voltages

A series RC circuit has R = 6000 Ω and C = 100 μF. The capacitor is initially uncharged: $v_C(0^-) = 0$ V. At t = 0, a constant 9 V source is connected and remains connected. Current is positive from the source through the resistor into the positive capacitor terminal. Use the following voltage balance and capacitor law:

$$v_S(t) = Ri(t) + v_C(t)$$

$$i(t) = C\frac{dv_C(t)}{dt}$$



a) Substitute the capacitor law into the voltage balance to derive the first-order ODE. Calculate the time constant, then apply the unilateral Laplace transform with the stated zero initial voltage to find $V_C(s)$.

b) Use partial fractions to obtain $v_C(t)$. Then find the charging current i(t) and evaluate both quantities at t = 0.6 s.

c) Find the capacitor voltage and charging current immediately after connection and after a long time. Explain each limiting result physically.

d) ⋆ Repeat (a)–(c) with initial capacitor voltage $v_C(0^-) = 3$ V; all other conditions are unchanged. Find $V_C(s)$, $v_C(t)$ and i(t), evaluate voltage and current at 0^+, 0.6 s and steady state, and explain what changes.

MATLAB verification. Use ode45 for initial capacitor voltages of 0 and 3 V. Keep the 9 V source connected.

Submit one figure with voltage and current in separate panels over 0–3 s. Overlay numerical and analytical responses and distinguish the two initial conditions. Report voltage and current at 0.6 s; check their initial and final values. Base MATLAB is sufficient.

Question 3 Repeated and complex poles

Invert the two transforms below for t ≥ 0. Use real-valued expressions and show the coefficient matching.

$$F(s) = \frac{2s+7}{(s+2)^2(s+5)},\quad H(s) = \frac{3s+9}{s^2+4s+13}$$

a) Find f(t) for F(s). Show why a repeated-pole term is necessary.

b) Find h(t) for H(s) by completing the square.

c) Check $f(0^+)$, $f'(0^+)$ and $h(0^+)$ directly and from high-frequency transform limits.

Question 4 When the final-value theorem works and when it fails

Compare two transforms for t ≥ 0. Check the poles of sY(s) before using the final-value theorem; verify each conclusion by finding the time response.

$$Y_A(s) = \frac{2}{s(s+2)}$$

a) A stable first-order system G(s) = 2/(s + 2) receives a unit step from rest. Use $Y_A(s)$ above to calculate its final value, then invert to check it.

$$Y_B(s) = \frac{1}{s^2+4}$$

b) A student substitutes s = 0 into $sY_B(s)$ and reports a final value of zero. Decide whether the theorem applies, invert $Y_B(s)$, and state whether a final value exists.

Method reference

Method source: Ogata, System Dynamics, 4th ed., Problem A-8-1, pp. 409–410 (first-order electrical response) and Chapter 2 (Laplace transforms). The RC values and the two charging cases are independently adapted. Page numbers refer to printed pages.

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